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Toward a sustainable post-pandemic growth

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ABSTRACT

This study investigates the long-run relationship between Research and Development (R&D) expenditure, Foreign Direct Investment (FDI) inflows, financial market capitalization, population, and Carbon Dioxide (CO₂) emissions across five ASEAN economies—Indonesia, Malaysia, Singapore, Thailand, and Vietnam—over 2000–2023. Building on the [Paramati et al. \(2021\)](#) framework rooted in the Impact, Population, Affluence, Technology (IPAT) theory of [Ehrlich and Holdren \(1971\)](#), the study applies Pedroni's panel cointegration test, the [Pesaran \(2007\)](#) CIPS unit-root test to address cross-sectional dependence, and Fully Modified OLS (FMOLS) estimation at both the panel and country-level. Results show that (i) population is the most consistently significant driver of CO₂ emissions across all five economies, (ii) R&D expenditure carries a negative but statistically insignificant coefficient, a result consistent with the region's structurally low R&D-to-Gross Domestic Product (GDP) ratios and lagged green-technology effects ([Yang et al., 2025](#); [Mehmood et al., 2022](#)), and (iii) FDI and financial-market effects are heterogeneous, supporting a Malaysia-Singapore green-finance interpretation alongside pollution-haven dynamics elsewhere. The findings highlight the need for ASEAN policymakers to scale R&D intensity, channel financial market deepening toward green sectors, and integrate demographic-driven emission pressures into national climate plans.

Keywords: ASEAN; FMOLS; cross-sectional dependence; Environmental Kuznets Curve; post-pandemic green recovery; financial market capitalization

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1. INTRODUCTION

The urgent need for sustainable economic recovery in the wake of the COVID-19 pandemic has brought environmental concerns to the forefront of policy discussions. As nations grapple with economic revival, there is growing recognition that green recovery strategies are essential for long-term growth. In this context, our study investigates the impact of economic policies on Carbon Dioxide (CO₂) emissions, aiming to identify pathways toward a more environmentally resilient future.

This study focuses specifically on five ASEAN economies, Indonesia, Malaysia, Singapore, Thailand, and Vietnam, a grouping that remains underexamined in the post-pandemic R&D–CO₂ literature despite their distinctive economic trajectories, heterogeneous green-policy frameworks, and growing carbon footprints. Prior empirical work on R&D and emissions has concentrated on the European Union (Paramati et al., 2021; Fernández et al., 2018), Organisation for Economic Co-operation and Development (OECD) members (Petrović & Lobanov, 2020), the BRICS bloc (Lee et al., 2021), and the UK (Shahbaz et al., 2020). Even where ASEAN is included, the dominant approach has been pooled panel estimation (Mehmood et al., 2022), which masks country-specific transmission mechanisms. To our knowledge, no prior study has produced a country-by-country Fully Modified OLS (FMOLS) decomposition of the R&D– Foreign Direct Investment (FDI)–financial-market–emissions nexus across these five ASEAN economies extending through the COVID-19 period (2020–2023). This study addresses that gap.

Specifically, this study examines how R&D expenditure, FDI inflows, financial-market capitalization, and population dynamics jointly influence CO₂ emissions across the selected ASEAN nations over 2000–2023. By applying Pedroni's panel cointegration framework alongside the Pesaran (2007) CIPS unit-root test and country-specific FMOLS estimation, the study contributes a long-run, country-by-country empirical decomposition that reveals heterogeneous policy transmission channels not captured by region-wide pooled estimates (e.g., Mehmood et al., 2022; Wei et al., 2023).

The ASEAN region merits dedicated empirical attention for three reasons. First, although Asia now accounts for roughly 45% of global R&D spending in 2024 (Bonaglia et al., 2025), ASEAN's R&D-to-GDP ratios remain well below the OECD average, Indonesia and Vietnam, despite recent strong upward trajectories, still spend under 0.5% of Gross Domestic Product (GDP) on R&D, raising open questions about whether the emission-reducing R&D effects documented in Europe (Paramati et al., 2021) translate to a region still in its industrialization phase. Second, post-pandemic recovery in Southeast Asia has been driven by manufacturing reshoring, semiconductor demand, and FDI inflows that may carry pollution-haven externalities (Luo et al., 2021; Oprea et al., 2025). Third, ASEAN's financial markets are deepening rapidly but unevenly, with Singapore's mature exchange contrasting Vietnam's nascent one, providing real cross-sectional variation in the financial-market–emissions channel. These structural features make ASEAN a uniquely informative laboratory for testing whether technology- and finance-driven decarbonization pathways generalize beyond advanced economies.

The study's novelty is threefold. First, it extends Paramati et al.'s (2021) Impact, Population, Affluence, Technology (IPAT)-based framework, originally applied to 25 European Union (EU) members, to a small-N, heterogeneous ASEAN panel that captures the post-pandemic period (2020–2023), an interval absent from most existing R&D–emissions studies. Second, it incorporates financial market capitalization as a proxy for green-capital allocation, building on Shahbaz et al. (2020) and Wang et al. (2020), and tests this channel separately for each ASEAN economy. Third, it provides the first country-level FMOLS decomposition across these five economies simultaneously, enabling direct policy comparison between, for example, Malaysia's mature capital-market signal and Vietnam's demographic-driven emissions profile.

Through empirical research and data-driven analysis, this study contributes substantively to the discourse on climate change economics, offering a robust foundation for evidence-based policy formulation and decision-making. The research outcomes have the potential to inform and shape national and international policy agendas, steering the global community towards a more sustainable and environmentally conscious economic trajectory in the aftermath of the pandemic.

2. LITERATURE REVIEW

The research conducted by [Huang et al. \(2022\)](#) delves into the intricate relationship between economic policies and CO2 emissions, with a specific focus on charting a path towards achieving a green economic recovery in the aftermath of the COVID-19 pandemic. Published in the esteemed journal *Climate Change Economics* in 2022, this study sheds light on the critical intersection of economic decision-making and environmental sustainability in the post-pandemic era. Studies conducted by [Huang et al. \(2022\)](#) had shown that regional efforts in the EU aimed at reducing pollution levels have been contrasting with individual member countries' unique policies to achieve a 10% reduction per country. The research contributes to the existing literature by developing a novel framework that investigates the link between macroeconomic policies, energy usage patterns, and CO2 emissions levels in EU countries from 1990 to 2016. The utilization of the second-generation Cross-Sectional-Autoregressive-Distributed Lag (CS-ARDL) panel data method in this study has enabled the identification of relationships between monetary policies and CO2 emissions, highlighting the impact of Growth Monetary Policy (GMP), Tightening Monetary Policy (TMP), environmental tax, carbon capture tax on environmental factors. Findings from the research indicate that GMP exacerbates the adverse effects of CO2 emissions, whereas TMP helps in reducing the harmful implications of CO2 emissions levels. Moreover, The Granger causality analysis authenticated our findings by confirming the one-way connection to CMP and EMP on carbon emissions, emphasizing the connectedness of economic policies and environmental concerns. In addition, Environmental Taxes have a negative and statistically significant relationship with CO2 emissions across all models. This suggests that higher environmental taxes are associated with lower CO2 emissions. The magnitude of the effect varies across models. Carbon Capture Taxes have a positive but statistically significant relationship with CO2 emissions in the OLS and DOLS models. This is counterintuitive, as we'd expect carbon capture taxes to reduce emissions. It could indicate a rebound effect (increased emissions due to other factors) or model misspecification. However, the effect is not significant in the FMOLS and PMG models. The study investigates the dual policy instruments (GMP and TMP) and their relationship with ecological factors provides a unique contribution to empirical analyses, shedding light on the importance of integrating environmental considerations into monetary policy decisions for achieving sustainable and green economic recovery in the aftermath of the COVID-19 pandemic.

In their study, [Petrović and Lobanov \(2020\)](#) explore the impact of Research and Development (R&D) expenditures on CO2 emissions across sixteen OECD countries. Their findings reveal a negative association between R&D spending and environmental degradation. Specifically, higher R&D expenditures tend to mitigate carbon emissions. However, several countries had different results. This study highlights the impact of R&D investments on CO2 emissions in 16 OECD countries between 1981 and 2014. While the expected average effect appears negative, the long-run impact can be positive in a significant percentage of cases. Interestingly, about 40% of countries do not follow this trend. Additionally, short-term analyses reveal varying effects; positive, negative, or neutral, over several years. These findings emphasize the need for empirical estimation rather than assuming a universally negative impact of R&D investments on emissions. Policymakers should focus on R&D activities that specifically reduce CO2 emissions or enable their capture, storage, and utilization. Furthermore, the study considers other relevant factors, such as GDP, which exhibits a positive long-term impact and aligns with the scale effect in the relationship between CO2 emissions and GDP.

The paper by [Fernández et al. \(2018\)](#) investigates the relationship between innovation, specifically R&D spending, and CO2 emissions. The authors analyze data from the European Union (EU-15), the United States, and China between 1990 and 2013 to determine if increased investment in innovation leads to a reduction in CO2 emissions. The study finds that R&D spending has a positive impact on reducing CO2 emissions in developed countries. This suggests that technological advancements and innovation can contribute to a more sustainable and environmentally friendly economy. The European Union demonstrates a more significant corrective effect compared to the United States, indicating regional differences in the effectiveness of innovation for emissions reduction. Furthermore, the paper highlights the role of energy consumption in CO2 emissions. As expected, increased energy consumption is linked

to higher emissions. However, the European Union (EU-15) again shows a lower impact, suggesting that its energy consumption is less polluting compared to the United States and China. The authors conclude that promoting R&D expenditure is crucial for combating climate change and achieving sustainable development.

The study conducted by [Shahbaz, et. al. \(2020\)](#), sheds light on the interplay between economic growth, financial development, R&D expenditures, and CO₂ emissions. The study situates itself within the context of two significant trends: the 4th industrial revolution and global decarbonization. The 4th industrial revolution, characterized by technological advancements like AI, IoT, and machine learning, has the potential to transform economies and societies. Importantly, if harnessed effectively, it could contribute to a cleaner environment. The authors explore how these trends intersect with the UK's commitment to achieving net-zero emissions by 2050. Methodologically, the researchers employ a bootstrapping bound testing approach to analyze short- and long-run relationships. Their dataset spans historical data from 1870 to 2017, allowing them to explore the dynamics over time.

The results indicate the existence of cointegration between CO₂ emissions and their determinants, suggesting that these variables are interconnected. Both financial development and energy consumption contribute to environmental degradation. Interestingly, R&D expenditures help reduce CO₂ emissions, aligning with the idea that innovation and research can drive sustainable practices. The relationship between economic growth and environmental effects follows an EKC pattern. Initially, as economies grow, emissions rise, but beyond a certain point, they decline due to improved technologies and policies. Financial development exhibits a U-shaped relationship with CO₂ emissions, implying that its impact is nonlinear. The study suggests using financial development and R&D expenditures as key tools to meet emissions targets in the fight against climate change. In summary, this research underscores the importance of innovation, sustainable finance, and strategic policy measures in achieving a net-zero carbon future for the UK.

The paper conducted by [Lee et al. \(2021\)](#) empirically analyzes sustainable relations between Inward Foreign Direct Investment (IFDI), Outward Foreign Direct Investment (OFDI), the R&D expenditure ratio and CO₂ emissions based on balanced panel data from the BRICS (namely, Brazil, Russia, India, China and South Africa) countries for the period 2003–2017. Commonly, the results confirm a negative effect of IFDI and a positive effect of OFDI on the R&D expenditure ratio, both with statistical significance. Further examination of the IFDI, OFDI and R&D impacts on CO₂ emissions was based on an assumption that innovation development mitigates environmental pollution. The research outcome revealed positive associations between IFDI and the R&D expenditure ratio with CO₂ emissions, showing the connection of investment growth-focused national economic strategies positive connections to CO₂ emissions. Based on these outcomes, they commend some strategies: the drafting of New Development Bank specific environment-friendly investment programs aimed at innovation activities and looking into further easing the green technologies from developed countries.

The study conducted by [Paramati, et al. \(2021\)](#) examines the long-run relationship between R&D investment and environmental sustainability in a panel of 25 EU member countries over a period of 17 years (1998–2014). They use robust and reliable econometric methods to capture the interactions between R&D investment on renewable energy consumption and CO₂ emissions. The findings confirm that the growth of R&D expenditures promotes renewable energy consumption and plays a significant role in reducing CO₂ emissions in the selected countries. Furthermore, the findings suggest that increasing the share of renewable energy consumption in the total energy mix reduces CO₂ emissions. These results suggest that EU policymakers should provide more financial and regulatory assistance to R&D activities, specifically in the energy sector, to promote low-carbon economies in this region. The empirical findings of their study confirm the presence of a significant long-run association among the variables: R&D, REC, and CO₂ emissions. The results also show that the growth of R&D expenditures positively contributes to increased renewable energy production and consumption and helps reduce CO₂ emissions in EU economies. The findings also indicate that increasing renewable energy consumption mitigates CO₂ emissions. These findings strongly suggest that by increasing R&D activities, EU countries not only promote renewable energy technologies and production but also help mitigate CO₂ emissions by

improving access to emission-controlling technologies. The major policy implications of this study are as follows. (i) Since the findings on long-run elasticities indicate that the growth of R&D activities increases REC and reduces CO₂ emissions, they argue that policymakers in EU countries should increase the funding allocated to R&D activities so that they can bring more innovations to new and existing sources of energy, particularly renewable energy, and introduce more emission-controlling technologies such as catalytic converters to reduce CO₂ emissions at source (of the automobile exhausts). These new innovations in the energy sector will greatly assist EU countries in further promoting the generation and use of renewable energy and combating the growth of CO₂ emissions. (ii) The results also show that the consumption of renewable energy decreases CO₂ emissions.

To achieve sustainable development, countries worldwide are choosing advanced technological capabilities and implementing environment-related innovations, including renewable energy. This offers an interesting opportunity for researchers to explore the unique determinants of carbon dioxide emissions. The N-11 economies are among the main emitters of carbon dioxide and energy consumption. Wang et al. (2020) analyses the dynamics of carbon emissions for N-11 countries from 1990 to 2017. They introduced innovative factors such as financial development, human capital, renewable energy consumption, and gross domestic product as determinants of CO₂ emissions. This study uses the second-generation panel cointegration method of 11 countries from 1990 to 2017 to assess the relationship between carbon emissions and country-specific variables. We found that, in the long run, financial development and economic growth are positively related to CO₂ emissions. In contrast, renewable energy consumption, technological innovation, and human capital are negatively related to CO₂ emissions in the N-11 economies. These findings remained robust when multiple econometric specifications were used. These findings have important implications and recommend the promotion of technological innovation and the use of renewable energy. This will help achieve the goals set by COP21.

Bimanatya and Widodo (2017) find heterogeneous causality between fossil fuel types and CO₂ in Indonesia (1965–2012) using VECM Granger causality, underscoring that Indonesia's emission dynamics are fuel-source-specific. This contextualizes why population rather than FDI dominates Indonesia's FMOLS results in the present study, as domestic fossil fuel consumption tied to demographic expansion drives much of the emissions trajectory.

Taken together, these three observations crystallize the contributions of the present study. First, the R&D-emissions literature is heavily skewed toward EU/OECD/BRICS samples, with ASEAN-specific evidence limited to pooled panel work (Mehmood et al., 2022; Wei et al., 2023) that conceals country-level heterogeneity. Second, the FDI and financial market channels are theoretically ambiguous and unlikely to operate uniformly across ASEAN's heterogeneous economies. Third, prior post-pandemic analyses for ASEAN tend to use ARDL/PMG frameworks that impose pooled long-run coefficients (Oprea et al., 2025; Feriansyah et al., 2022), whereas a country-specific FMOLS approach, followed in this study, permits idiosyncratic long-run elasticities while retaining the bias-correction advantages of cointegration estimation. The above synthesis justifies the empirical model specified in the next section.

3. DATA AND EMPIRICAL METHODOLOGY

Sample selection reflects data availability constraints rather than theoretical design. The five countries (Indonesia, Malaysia, Singapore, Thailand, Vietnam) are those for which the full vector of variables, CO₂ emissions, FDI inflows, GDP per capita, population, R&D expenditure (% of GDP), and market capitalization (% of GDP), is consistently reported across the full 2000–2023 panel period in the World Development Indicators database. Consistent R&D and market capitalization series are unavailable for most other ASEAN members over this window. These five economies collectively account for the majority of ASEAN's GDP and span a meaningful spectrum of industrialization stages, from Singapore's high-income service-led economy to Vietnam's rapidly industrializing manufacturing base, providing meaningful cross-sectional variation despite the small N.

This study developed an adjusted model from Paramati et al. (2021) to investigate the impact of R&D expenditure on CO₂ emissions, we use the environmental theoretical model (Ehrlich & Holdren

(1971) mentioned in [Paramati et al. \(2021\)](#) to determine the drivers of CO2 emissions. This theoretical model is built based on the association among population, income, technology, and environmental impact. To account for various other potential drivers of emissions, we base our empirical model on [Paramati et al. \(2021\)](#), as described below.

$$CDE_{it} = f(POP_{it}, PI_{it}, R\&D_{it}, FDI_{it}, FM_{it}, \dots)$$

This equation aims to identify the role of R&D in reducing the CO2 emissions. where CDE, PI, R&D, FDI, and FM denote CO2 emissions, per capita income, R&D expenditure, Foreign Direct Investment (FDI), and financial markets, respectively. CDE for total CO2 emissions (kt); FDI for FDI, net inflows (% of GDP); and PI for GDP per capita (constant 2015 U.S.\$); POP for the total population; R&D for R&D expenditure (% of GDP); and FM for market capitalization (% of GDP).

Variable selection follows the IPAT framework ([Ehrlich & Holdren, 1971](#)), which attributes environmental impact to population scale (P), economic affluence (A), and technology (T). Population (POP) captures the scale of human activity; per-capita income (PI) represents affluence and enables testing of the EKC hypothesis through a non-linear specification; and R&D expenditure proxies for technology and is the primary variable of interest. Beyond the IPAT tripod, two additional regressors are theoretically motivated as follows. FDI inflows are included because, in ASEAN, foreign investment is the principal channel through which cleaner technologies (or, alternatively, pollution-intensive production) enter the domestic economy, operationalizing the pollution halo/pollution haven debate. Financial market capitalization is introduced as a proxy for financial deepening and capital allocation capacity, extending the broader financial development perspective discussed by [Shahbaz et al. \(2020\)](#) and [Wang et al. \(2020\)](#). The absence of severe multicollinearity among variables is confirmed through diagnostics reported in Annex (see [Table 1](#)).

Table 1. Multicollinearity Diagnostic (VIF)

Variable	Coef. Variance	Uncentered VIF
D(FDI_INFLOWS)	202,717.1	1.588
D(POP)	9.89E-07	1.020
D(RD)	2.01E+08	1.250
D(MARKETCAP)	3,293.6	1.606

All Variance Inflation Factors were well below the conventional threshold of 5 (range 1.02–1.61), indicating no multicollinearity concern. Each variable contributes an independent variation to the regression, supporting the joint inclusion of POP, R&D, FDI, and MARKETCAP in the cointegrating equation.

ASEAN economies are tightly interconnected through trade, FDI, and shared macro shocks (notably COVID-19), making Cross-Sectional Dependence (CSD) a structural feature of the panel, rather than a technical nuisance. First-generation unit root tests (LLC or [Levin, Lin, and Chu's \(2002\)](#) test and IPS or [Im, Pesaran, and Shin's \(2003\)](#) test) assume cross-sectional independence; relying on them alone risks over-rejection of the unit root null. Therefore, we complement them with [Pesaran's \(2007\)](#) CIPS test, a second-generation procedure that absorbs common factor dynamics through cross-sectionally augmented regressions. The Annex confirms convergence between the two generations: for D(MARKETCAP), the CIPS statistic is -4.638 ($p < 0.01$) and Truncated CIPS -3.897 ($p < 0.01$), well beyond the 1% critical value of -2.78 ; for D(POP), the Truncated CIPS is -2.794 ($p < 0.01$) against a 1% critical value of -2.594 ; for D(Y), the CIPS is -2.629 ($p < 0.01$); and for FDI_INFLOWS, the CIPS test rejects at the 10% level (-2.293), consistent with the LLC/IPS rejection of the unit root null at the 1% level for FDI_INFLOWS and RD tested in levels. The convergence of first- and second-generation tests on first-difference stationarity, therefore, robustifies the I(1) characterization of the data and validates the subsequent Pedroni cointegration and FMOLS steps despite the presence of cross-sectional dependence.

Long-run cointegration was tested using [Pedroni's \(1999\)](#) and [Pedroni's \(2004\)](#) panel cointegration framework. All variables were found to be integrated of order one $I(1)$ and were entered in the first differences in the cointegrating regression.

This study employs FMOLS because the variables are integrated of order one $I(1)$ and cointegrated, meaning that standard OLS produces biased estimates due to endogeneity and serial correlation inherent in non-stationary panels. FMOLS corrects for both problems through non-parametric adjustments to the long-run covariance matrix, without requiring a pre-specified lag structure, a critical advantage given this panel's short time dimension. Thus, FMOLS provides consistent, bias-corrected long-run elasticity estimates under the constraints of this dataset.

The five-country sample reflects a binding data constraint, not a theoretical choice: consistent R&D and market capitalization series are unavailable for most other ASEAN members over the full 2000–2023 window. This limitation has genuine methodological consequences. With only five cross-sections, the panel has limited statistical power to cleanly separate country-specific idiosyncrasies from broader regional dynamics, and pooled-country coefficients inevitably average over substantial structural heterogeneity between, for example, Singapore and Vietnam. The five included countries are also ASEAN's most developed; thus, the findings should not be uncritically extended to lower-income members (Cambodia, Laos, and Myanmar), whose institutional and innovation systems differ markedly. Therefore, the results are best read as indicative long-run evidence for this specific grouping during the 2000–2023 window, with future research extending coverage as regional data infrastructure improves. We explore the long-run association among these models using the [Pedroni \(1999\)](#) and [Pedroni \(2004\)](#) panel cointegration framework mentioned in [Paramati et al. \(2021\)](#).

4. RESULTS AND DISCUSSION

4.1. Panel cointegration framework

For more details, see [Table 2](#) and [Table 3](#).

Table 2. Alternative Hypothesis: Common AR Coefs. (Within-Dimension)

	Statistic	Prob.	Weighted Statistic	Prob.
Panel v-Statistic	-1.06845	0.8573	-2.29785	0.9892
Panel rho-Statistic	-1.09162	0.1375	0.281544	0.6109
Panel PP-Statistic	-8.2976	0.0000**	-8.86259	0.0000**
Panel ADF-Statistic	-7.68459	0.0000**	-4.12603	0.0000**

Table 3. Alternative Hypothesis: Individual AR Coefs. (Between-Dimension)

	Statistic	Prob.
Group rho-Statistic	1.065127	0.8566
Group PP-Statistic	-10.8732	0.0000**
Group ADF-Statistic	-5.53825	0.0000**

The findings of the panel cointegration framework show that out of the test's seven statistics, two statistics under the within-dimension and two statistics under the between-dimension statistics are statistically significant. This means that there is a considerable long-run relationship between the variables under consideration.

We followed the method described by [Paramati et al. \(2021\)](#), who used the cointegrating regression FMOLS in all countries and specified countries. The results are described as follows:

Table 4. All Countries (Cointegrating Regression FMOLS)

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(FDI_INFLOWS)	406.2972	450.2411	0.902399	0.3695
D(POP)	0.005733	0.000995	5.764072	0.00**
D(RD)	-5411.77	14170.75	-0.3819	0.7035
D(MARKETCAP)	-39.471	57.38996	-0.68777	0.4935

Source: Author's calculation based on World Development Indicators data.

The FMOLS results in Table 4 show population growth as the most consistently significant determinant ($\beta = 0.005733$, $p < 0.01$), a result corroborated by Effendi (2026), whose ASEAN+3 estimates confirm population elasticities close to or above unity, consistent with demographic scale effects being a dominant structural correlate of aggregate emissions.

Table 5. Specified Analysis Per Countries (Cointegrating Regression FMOLS)

Country	Variable	Coefficient	Std. Error	t-Statistic	Prob.
Indonesia	D(FDI_INFLOWS)	-10.0952	4881.486	-0.00207	0.9984
	D(POP)	0.004668	0.001892	2.46741	0.0261**
	D(RD)	-6081.44	213705.4	-0.02846	0.9777
	D(MARKETCAP)	317.0707	452.7372	0.700341	0.4944
Malaysia	D(FDI_INFLOWS)	2121.961	770.1413	2.755288	0.0147**
	D(POP)	0.014372	0.002612	5.501711	0.0001**
	D(RD)	-17997.5	12309.89	-1.46204	0.1644
	D(MARKETCAP)	-142.93	44.04514	-3.24507	0.0054**
Singapore	D(FDI_INFLOWS)	117.7192	54.32429	2.166972	0.0467**
	D(POP)	0.004101	0.002681	1.529832	0.1469
	D(RD)	-2677.18	1989.517	-1.34565	0.1984
	D(MARKETCAP)	-16.9374	8.518715	-1.98826	0.0653**
Thailand	D(FDI_INFLOWS)	1897.72	856.0207	2.216909	0.0425**
	D(POP)	0.014357	0.003374	4.254437	0.0007**
	D(RD)	-24581.7	14466.31	-1.69924	0.1099
	D(MARKETCAP)	-27.7464	66.69197	-0.41604	0.6833
Vietnam	D(FDI_INFLOWS)	2828.159	8021.16	0.352587	0.7348
	D(POP)	0.027732	0.00626	4.430183	0.003**
	D(RD)	-165785	100673.4	-1.64676	0.1436
	D(MARKETCAP)	306.4117	783.8324	0.390915	0.7075

Source: Processed from primary data

The consistent insignificance of R&D across all five ASEAN countries warrants careful interpretation rather than dismissal as a null finding. Four mechanisms underlie this result. First, ASEAN economies have historically maintained R&D expenditures well below the OECD average of approximately 2.7% of GDP. Even in 2024, most ASEAN-5 members invest under 1% of GDP in R&D (Bonaglia et al., 2025), an intensity that may be too low to register measurable abatement effects at the

national-emissions level. Second, R&D investment typically operates through lagged channels; green technology developed in year t reduces emissions in years $t+k$, but a single-period cointegrating coefficient predominantly captures contemporaneous effects. [Yang et al. \(2025\)](#) explicitly document a delayed but beneficial long-term influence of environmental innovation in ASEAN, supporting this lag-effect interpretation. Third, the composition of R&D spending in these countries may be weighted toward industrial-process innovation, ICT, and biomedical research rather than clean energy and emission-control technologies, limiting environmental traction. Fourth, weak green technology transfer through FDI channels may further dilute the R&D–emissions linkage. This pattern is consistent with [Petrović and Lobanov \(2020\)](#), who found that roughly 40% of OECD countries exhibit non-negative R&D–emission relationships, underscoring that R&D's environmental effectiveness is conditional on the national innovation system and regulatory incentive structure (see [Table 5](#)).

The country-specific results reveal substantively heterogeneous emission drivers with distinct policy implications. In Indonesia, only population growth is significant, consistent with its position as ASEAN's most populous economy, undergoing rapid urbanization. The insignificance of FDI and financial market variables suggests that domestic consumption- and demographic-driven emissions dominate externally financed production in this case. In Malaysia, FDI, population, and market capitalization are all statistically significant; critically, the market capitalization coefficient is negative (-142.93 , $p = 0.0054$), pointing to a green-allocation interpretation in which Malaysia's relatively mature capital market channels investment toward less carbon-intensive listed sectors, consistent with the U-shaped financial-development mechanism in [Shahbaz et al. \(2020\)](#). Singapore shows a positive significant FDI coefficient alongside a marginally significant negative market cap effect, plausibly reflecting the carbon footprint of data centers, financial-sector infrastructure, and shipping/logistics FDI on the one hand, and green-finance capital reallocation on the other. Thailand mirrors Malaysia's population–FDI dynamics but with insignificant market-cap effects, suggesting that its capital market has not yet reached the depth at which green-allocation effects materialize empirically. Vietnam's uniquely large population coefficient (0.0277 , the highest of any country) reflects its exceptionally rapid demographic and industrial expansion from 2000 to 2023. The insignificance of FDI and market cap variables likely reflects the still-shallow market structure and short post-equitization sample. Taken together, the variation across countries reinforces the argument for country-specific rather than pooled estimation.

This research aligns with [Effendi \(2026\)](#), as many ASEAN+3 economies, such as Cambodia, China, Indonesia, Lao PDR, Malaysia, Myanmar, the Philippines, Thailand, Timor-Leste, and Vietnam, sit below the turning point range. For these economies, further income gains could increase emissions unless offset by gains in energy efficiency, structural change, and mitigation policies. For economies near or beyond their estimated turning points, growth risks arise from structural rigidities and policy execution rather than from scale effects. In these economies, insufficient progress in energy efficiency, fuel switching, or technological upgrading can result in higher production costs, reduced competitiveness, and vulnerability to energy price shocks. Delayed transition in hard-to-abate sectors may also hinder potential growth by locking in carbon-intensive capital and slowing investment in emerging low-carbon industries.

Aligned with [Paramati et al. \(2021\)](#), our results provide direct evidence that growth in R&D and well-functioning financial markets can reduce CO₂ emissions, but only conditionally. The negative significant market-cap coefficient for Malaysia (-142.93 , $p = 0.0054$) and marginally significant negative coefficient for Singapore (-16.94 , $p = 0.0653$) directly support the green-allocation interpretation: in financially deeper ASEAN markets, capital appears to migrate toward less carbon-intensive listed sectors, consistent with growing environmental, social, and governance-aligned investor preferences. By contrast, Indonesia's positive (insignificant) market-cap coefficient implies that its shallower market channels capital predominantly into industrial expansion. This empirical contrast between countries with different financial market depths corroborates [Shahbaz et al.'s \(2020\)](#) U-shaped mechanism within a single cross-section.

[Hanif and Gago-de-Santos \(2017\)](#) in their 2017 study raised an important point about how developing economies might experience the Environmental Kuznets Curve differently than wealthier nations. They suggested that while a turning point in emissions could eventually be reached, the path was far from guaranteed. Rising populations, inflation, and fragile income growth can all slow progress,

meaning that instead of the expected inverted U-shaped curve, economic growth may continue to increase emissions. Their warning reminds us that growth without stability and foresight risks locking countries into environmentally damaging trajectories.

Hanif and Gago-de-Santos (2017) caution about EKC trajectories in developing economies is directly evidenced by the present findings. Across all five ASEAN countries, population growth emerges as the most consistently significant predictor of CO₂ emissions in our FMOLS estimates, a result that aligns precisely with the warning that demographic pressure can override the emission-dampening effects of income growth and technological investment. The absence of a significant R&D effect in our pooled and country-specific regressions further suggests that these economies have not yet reached the threshold of innovation intensity at which technological progress translates into measurable decarbonization, a state consistent with the pre-turning-point phase of the EKC. The Annex EKC plots confirm this empirically: of the five countries, only Singapore exhibits a clear inverse-U pattern in the GDP-per-capita–CO₂ plot, signalling that it alone has approached the post-industrial knowledge-economy phase, while Indonesia, Malaysia, Thailand, and Vietnam all remain on the rising portion of the curve. Policymakers across the four pre-turning-point economies should therefore prioritize raising R&D-to-GDP intensity as a structural precondition for harnessing the emission-reducing potential of innovation, rather than treating current R&D levels as adequate.

Each policy implication can now be tied to a specific empirical finding from FMOLS estimates. (i) The significant negative market-cap coefficient in Malaysia and the marginally significant negative coefficient in Singapore motivate the sustained development of ASEAN's listed green-finance instruments, green bonds, sustainability-linked equities, and ESG-aligned mutual funds, particularly in Indonesia, Thailand, and Vietnam, where this channel has not yet emerged empirically. (ii) The universally insignificant but negative R&D coefficient implies that current ASEAN R&D intensity is below the threshold at which emission-reducing technology effects become statistically detectable; a coordinated regional commitment to lift R&D-to-GDP above 1%, particularly directed toward clean energy, emission control, and energy efficiency technologies, is therefore the empirically grounded policy priority. (iii) The positive and significant population coefficient in four of five countries implies that demographic and urbanization pressures must be addressed through targeted urban planning, transport electrification, and clean cooking fuel interventions rather than being left as residual scale effects. (iv) The positive significant FDI coefficient in Malaysia, Singapore, and Thailand calls for FDI-screening frameworks that filter for clean-technology content, operationalizing the pollution-halo channel while restraining the pollution-haven channel evident in industrial-sector inflows.

In summary, achieving the EKC turning point for ASEAN-4 (Indonesia, Malaysia, Thailand, and Vietnam) requires coordinated action on three fronts identified by our results: raising R&D intensity to activate the innovation-decarbonization channel, deepening green financial markets to trigger the capital-reallocation effect already visible in Malaysia, and integrating demographic pressure into land-use and urban emission policies.

5. CONCLUSION

This study applied Pedroni panel cointegration and country-specific FMOLS estimation to investigate the long-run drivers of CO₂ emissions across five ASEAN economies, Indonesia, Malaysia, Singapore, Thailand, and Vietnam, over 2000–2023, with an explicit treatment of cross-sectional dependence through the Pesaran CIPS test. Four conclusions were drawn. First, population is the most consistently significant emission driver across the region, validating Hanif and Gago-de-Santos (2017) warning about demographic pressures in pre-turning-point EKC economies. Second, R&D expenditure has a negative but statistically insignificant effect across all countries, a result that reflects the region's structurally low R&D intensity, lagged green technology transmission, and composition effects rather than an absence of innovation-driven decarbonization potential. Third, financial market capitalization exerts a significant emission-reducing effect in Malaysia and a marginal effect in Singapore, supporting the green-allocation interpretation of mature capital markets while reminding policymakers that this channel has not

yet been activated in the region's shallower markets. Fourth, FDI shows a heterogeneous pattern consistent with pollution-haven dynamics in economies with significant manufacturing and FDI-intensive sectors (Malaysia, Singapore, and Thailand) but is statistically inert in Indonesia and Vietnam. Policy priorities that raise R&D-to-GDP intensity, deepen green-aligned listed markets, integrate demographic and land-use management into climate plans, and screen FDI for clean-technology content follow directly from these empirical results. The study's limitations include the five-country sample size, the 24-year window that may understate longer-cycle R&D effects, and the use of national CO₂ totals that do not distinguish between sectoral sources. These are appropriate directions for future research.

Ethical Approval

Ethical approval was not required because this study used secondary macroeconomic data and did not involve human participants or personal data.

Informed Consent Statement

Informed consent was not applicable because the study did not involve human participants.

Authors' Contributions

The author solely conceptualized the study, designed the methodology, collected and analyzed the data, wrote the original manuscript, reviewed and edited the manuscript, and approved the final version for publication.

Disclosure Statement

No potential conflict of interest was reported by the author(s).

Data Availability Statement

The data used in this study are publicly available from the World Development Indicators and other sources cited in the article. Processed datasets are available from the corresponding author upon reasonable request.

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